GENERATIVE MODELING OF RUPTURE PLANE FROM FINITE-SOURCE RUPTURE PARAMETERS

Manish Nayak
CE22B069

Guided by
Prof. STG Raghukanth

INTRODUCTION

Accurate slip prediction and slip distribution is crucial for:

- Uncertainty Estimation For Early Disaster Management
- Seismic Hazard and Stress-Transfer Analysis
- Earthquake Engineering & Infrastructure Design
- Site-Specific Seismic Response Studies

DRAWBACKS OF PREVIOUS WORKS

1. High Computational Cost

Physics-based simulations (dynamic/kinematic) require solving frictional PDEs → extremely slow.
Refs: Graves & Pitarka (2010); Richards-Dinger & Dieterich (2012 – RSQSim)

2. Limited Data Availability and Poor Generalizability

Global finite-fault datasets (SRCMOD) have only ~250–300 events, which is insufficient for standard deep learning to generalize properly.
Ref: Mai & Thingbaijam (2014)

3.No current model jointly uses detailed fault geometry + source parameters to generate full 2D rupture plane.

OBJECTIVES

- To develop a generative deep learning model that predicts 2D rupture plane using only fault geometry and source parameters.
- To provide fast and generalizable slip values and spread of rupture planes.

↓

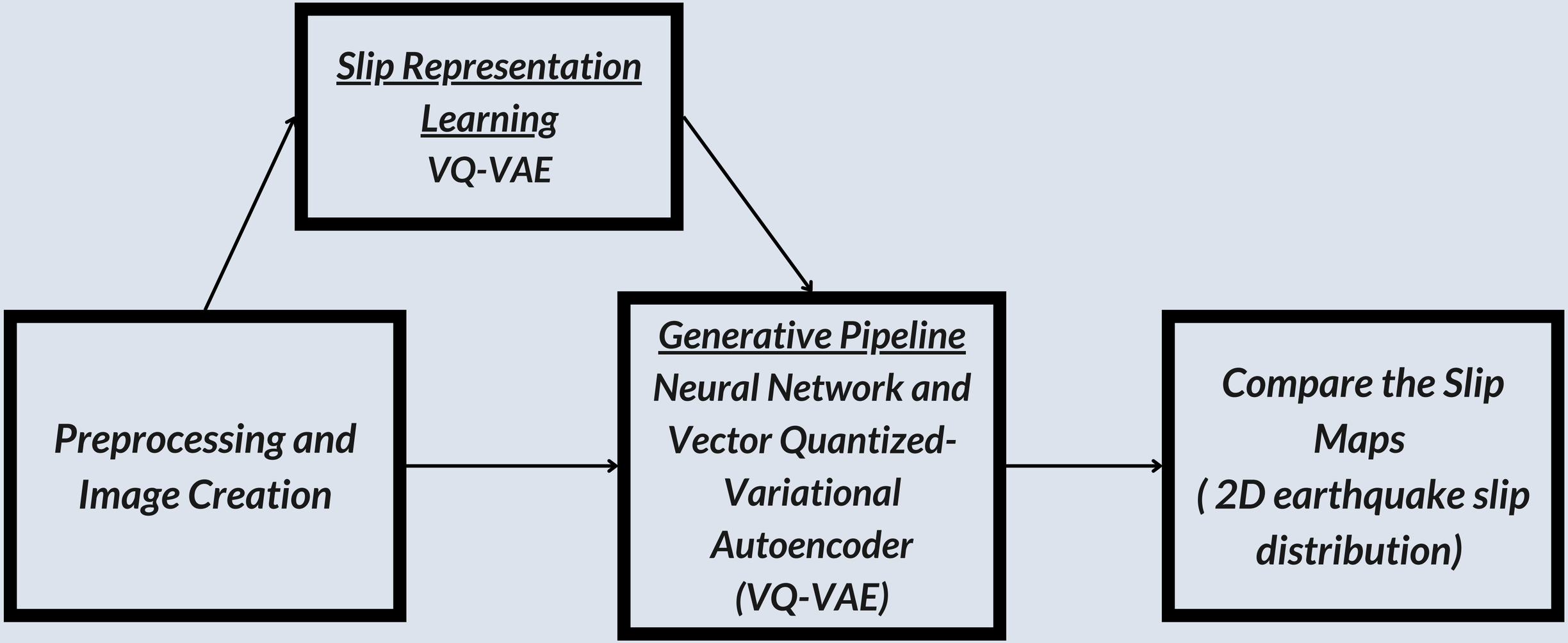
Usage of a novel pipeline

Combines the underlying distribution learning ability of VQ-VAEs with simpler mapping ability of Neural Networks and Generalizability-focused Loss function

↓

Neural Network and Vector Quantized-Variational Autoencoder

METHODOLOGY



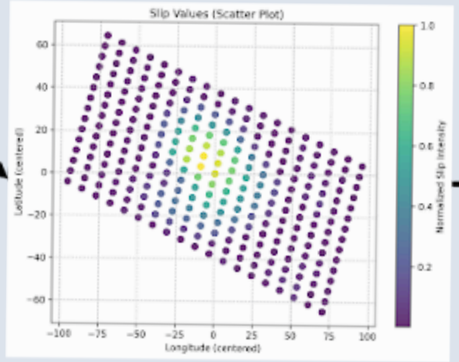
PREPROCESSING AND IMAGE CREATION

Original *.fsp* files obtained from numerical simulations.

Latitude	Longitude	Depth	DIP	....	M_w	STRK
37.78	-122.51	10.00	90.0	....	7.91	325
....	....	....	....	....		
37.90	26.82	10.04	48.0	....	7.05	276.0

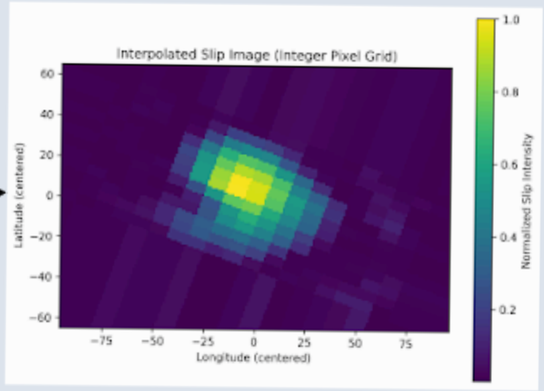
Original dataset extracted

Choose **15** input parameters.



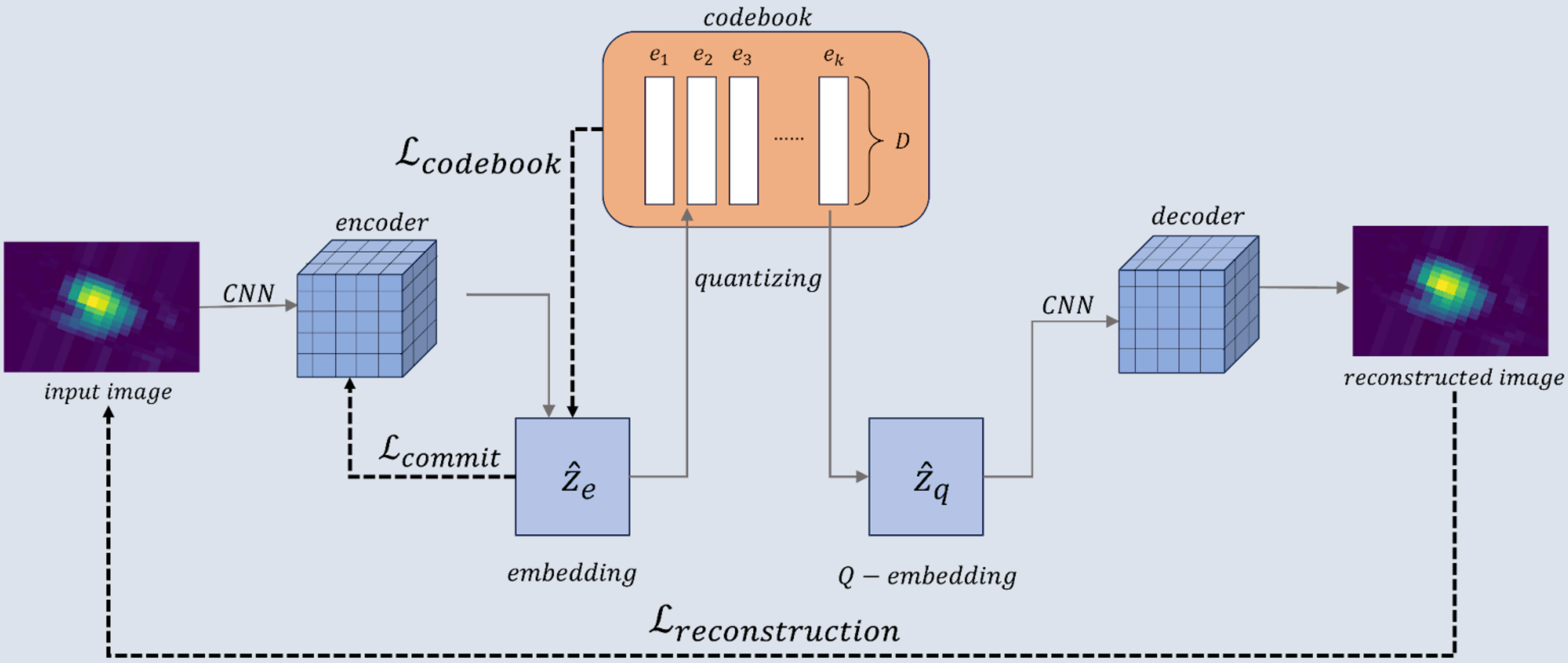
Slip values (sparse form)

Bicubic interpolation

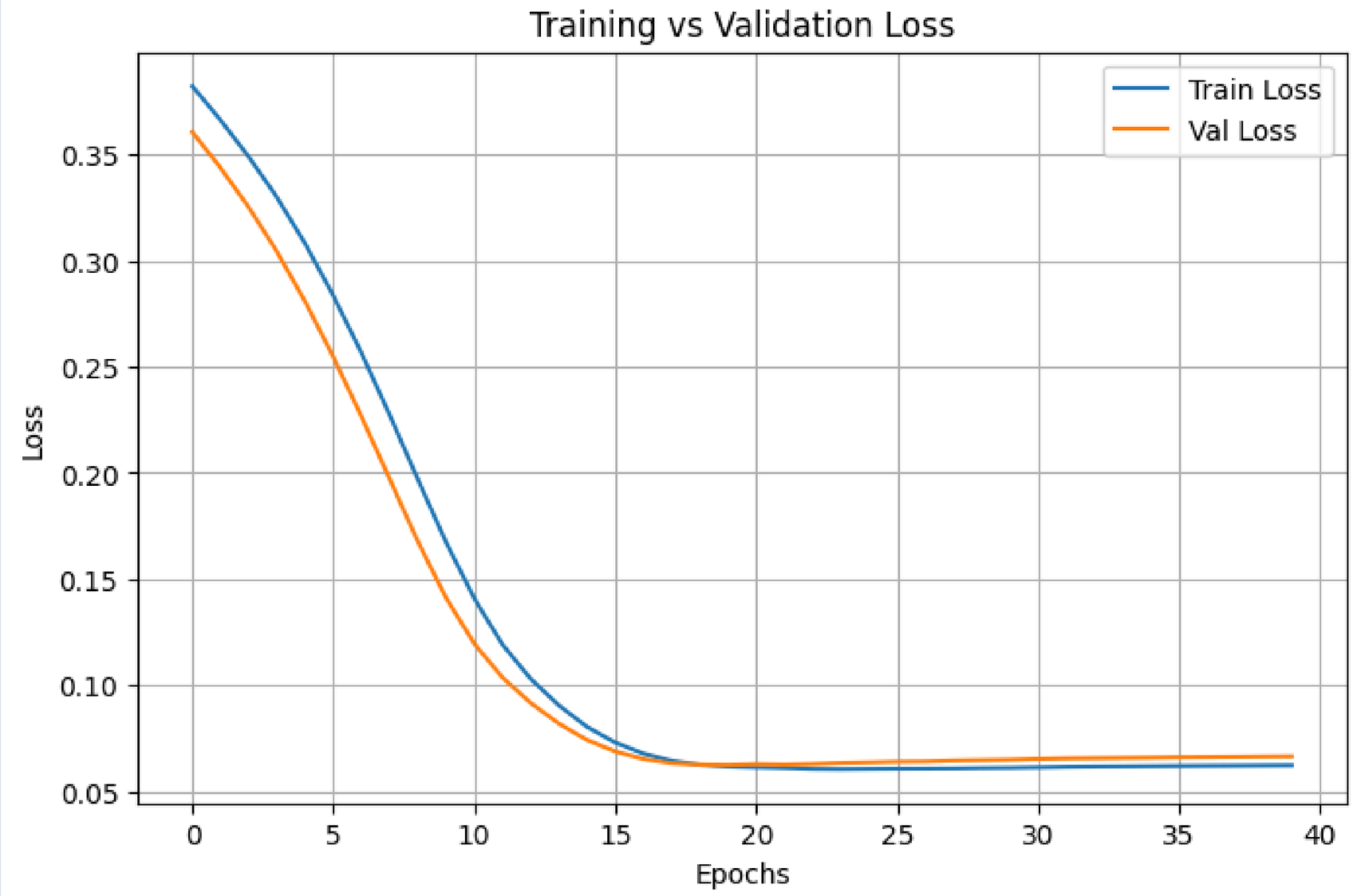


Interpolated image

SLIP REPRESENTATION LEARNING (VQ-VAE)



ERROR PLOT FOR SLIP REPRESENTATION LEARNING PIPELINE



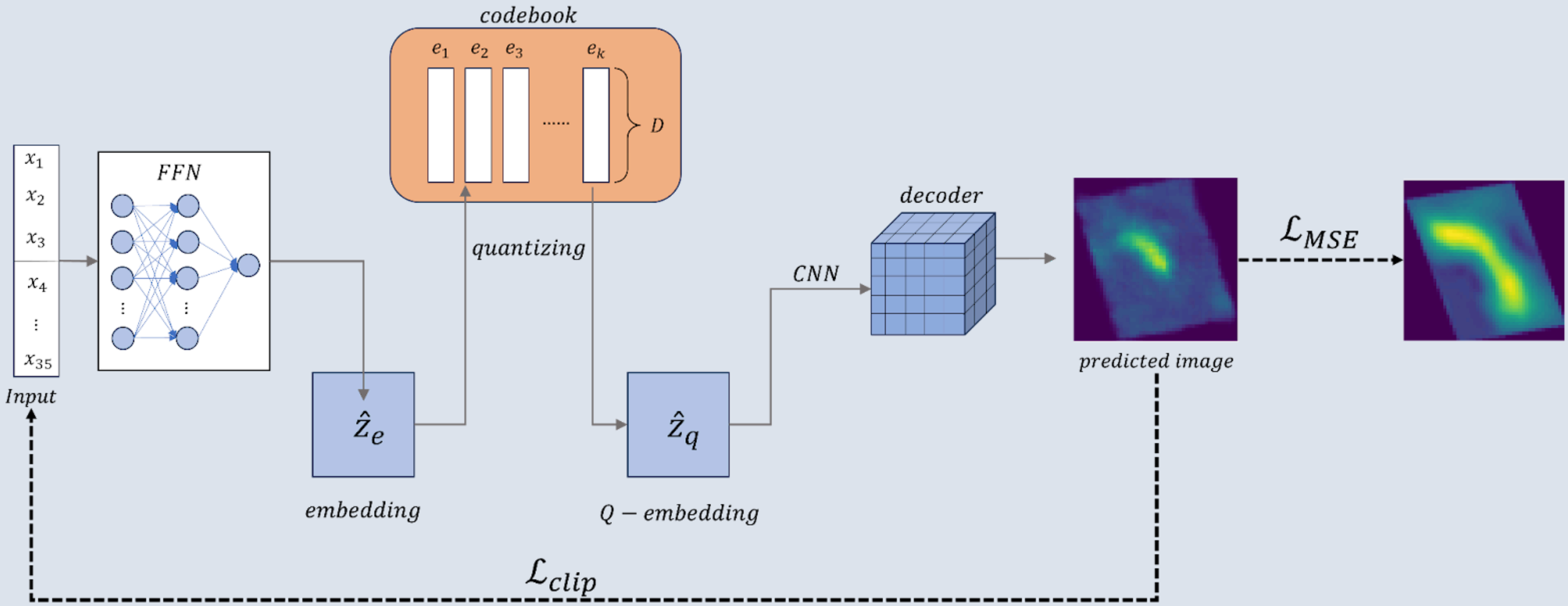
INPUT FEATURES (SRCMOD DATASET)

FEATURES	JUSTIFICATION FOR USE
Moment Magnitude (Mw)	Controls the total seismic moment and governs the overall slip amplitude, directly influencing rupture size and ground-motion severity.
Latitude (LAT) & Longitude (LON)	Encode regional tectonic setting and geological context, which affect rupture characteristics and spatial variability in slip.
Strike Angle (STRK), Dip Angle (DIP), Rake Angle (RAKE)	Describe the fault's geometry and slip mechanism. These parameters influence rupture kinematics, shear direction, and the overall spatial organization of slip. • Strike → Orientation of the fault relative to North. • Dip → Steepness of the fault plane. • Rake → Direction of Motion on the fault plane.

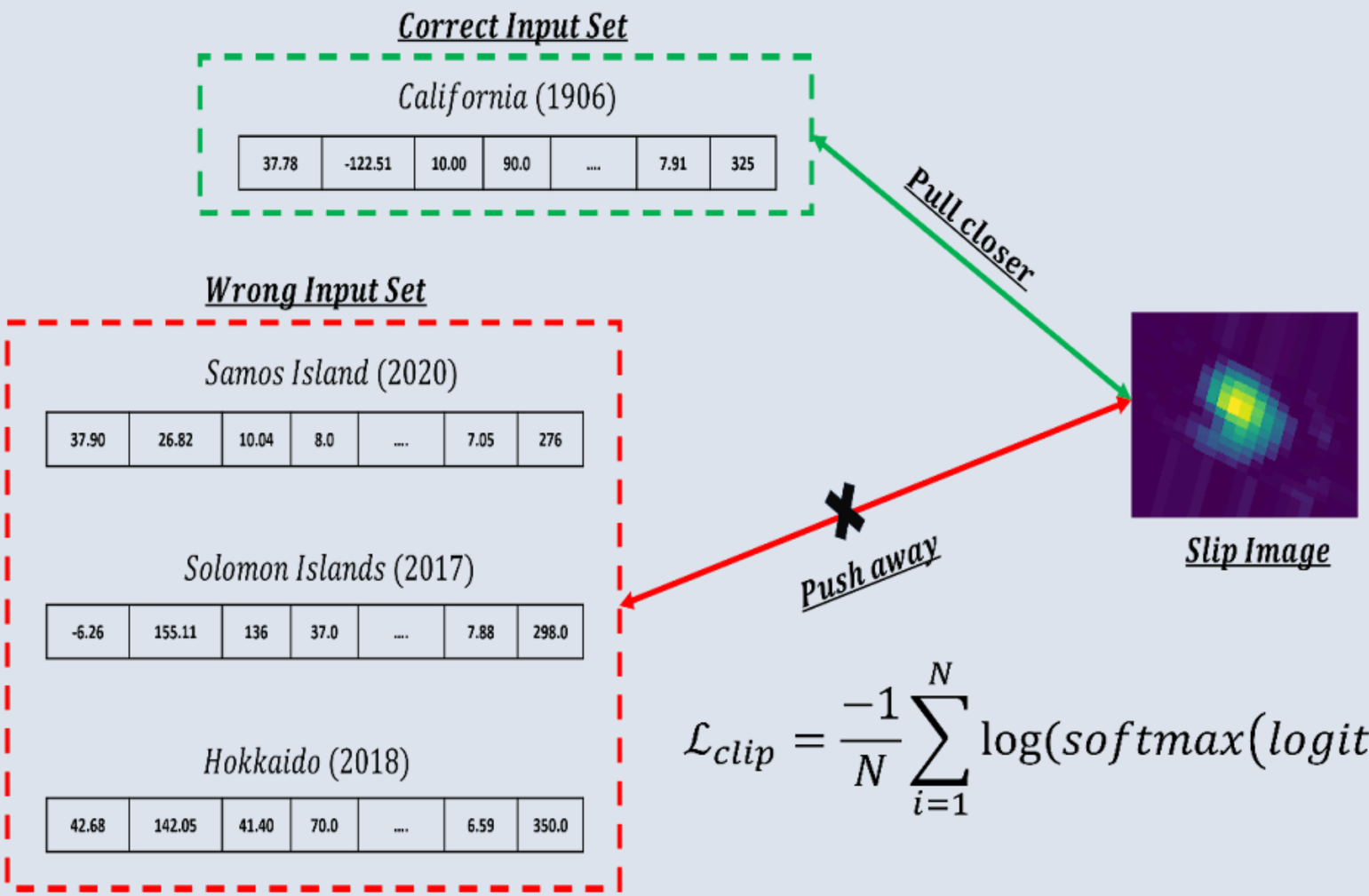
INPUT FEATURES (SRCMOD DATASET)

Fault Depth (DEP)	Depth influences lithostatic stress and frictional conditions; deeper regions typically exhibit lower slip due to higher confining pressure.
Depth to Top of Fault (Htop)	Controls how close the rupture reaches the surface; shallow ruptures can produce larger surface slip and stronger shaking relevant to infrastructure performance.
Fault Length (LEN_f) & Fault Width (WID)	Define the geometric dimensions and total rupture area, constraining the maximum possible moment release and slip distribution pattern.
Hypocenter Position Along Strike (HypX) & Hypocenter Position Down Dip (HypZ)	Nucleation location determines how rupture propagates, affecting where slip concentrates and how asperities develop across the fault plane.

SLIP GENERATION PIPELINE

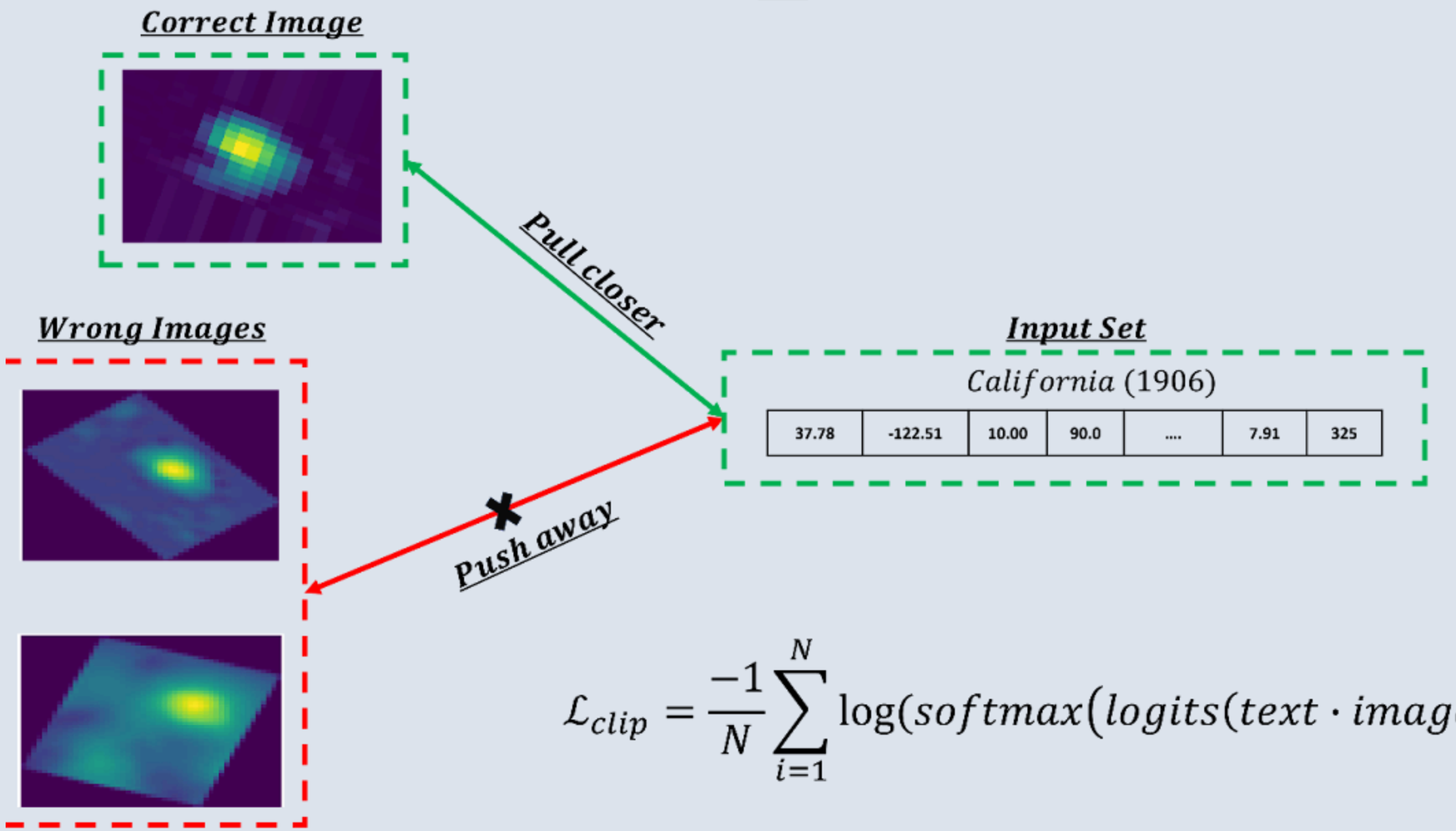


CLIP LOSS-I



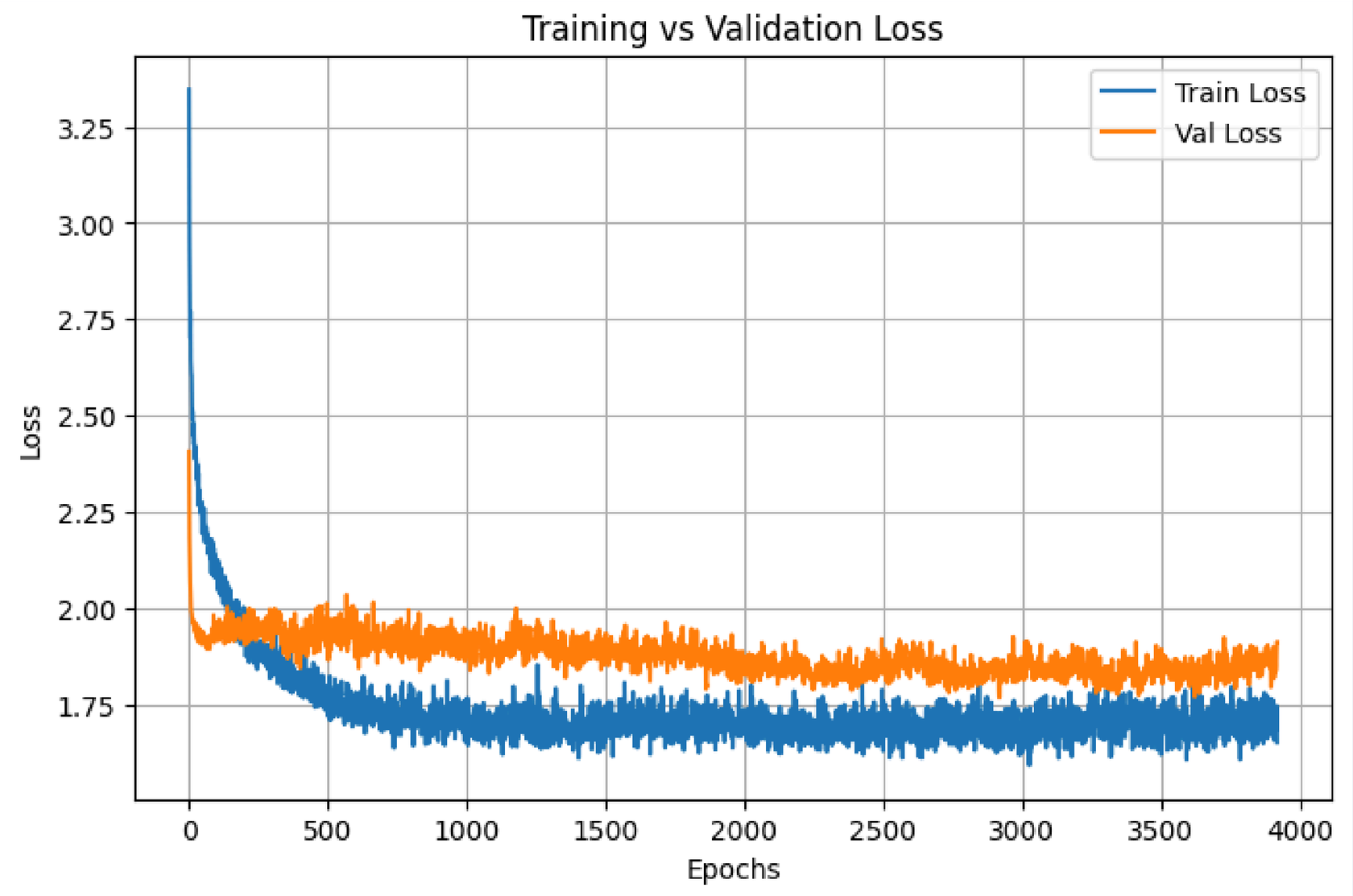
$$\mathcal{L}_{clip} = \frac{-1}{N}\sum_{i=1}^{N} \log(softmax(logits(image \cdot text)))$$

CLIP LOSS+II



$$\mathcal{L}_{clip} = \frac{-1}{N}\sum_{i=1}^{N} \log(softmax(logits(text \cdot image)))$$

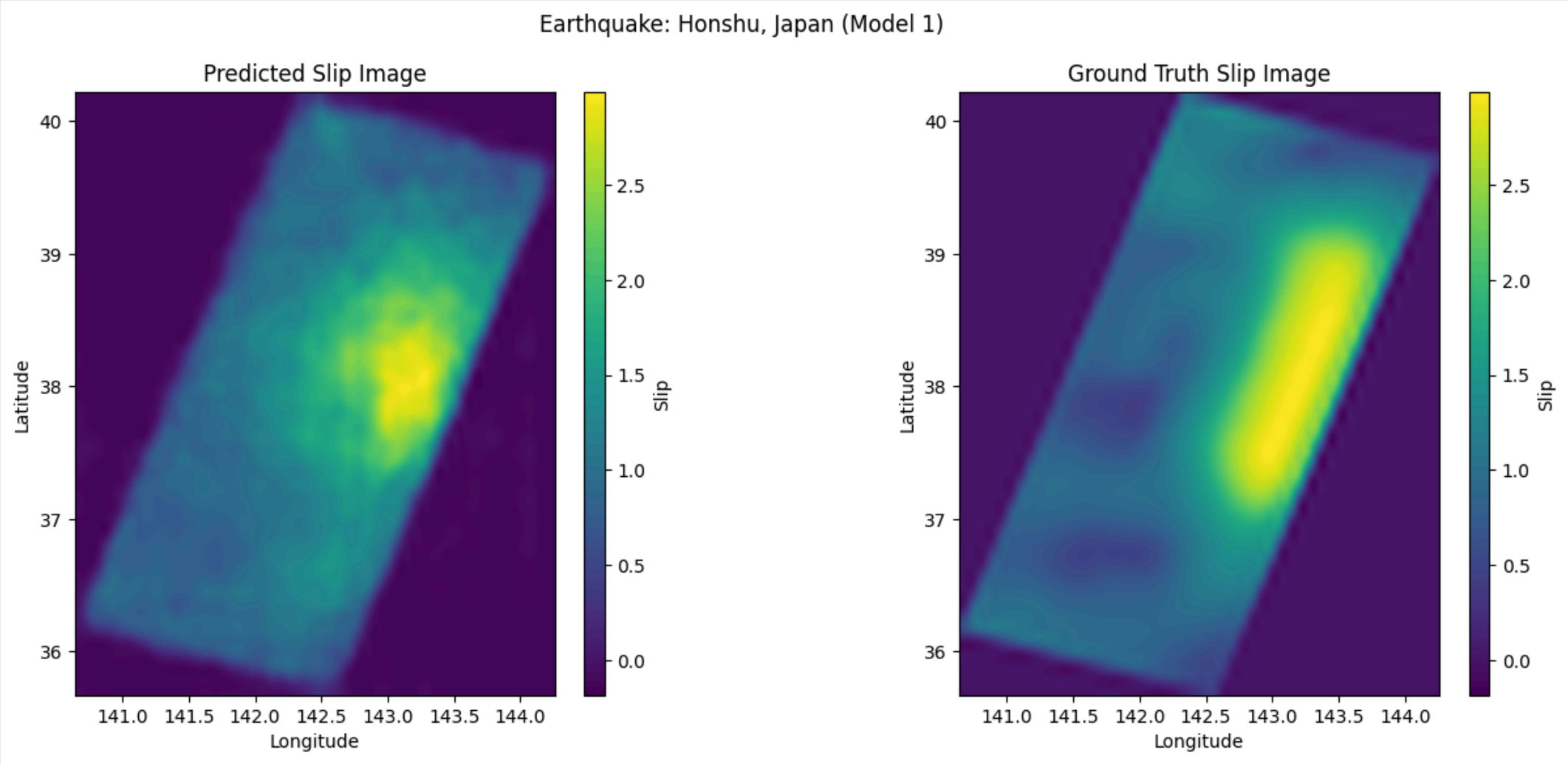
ERROR PLOT FOR SLIP GENERATION PIPELINE



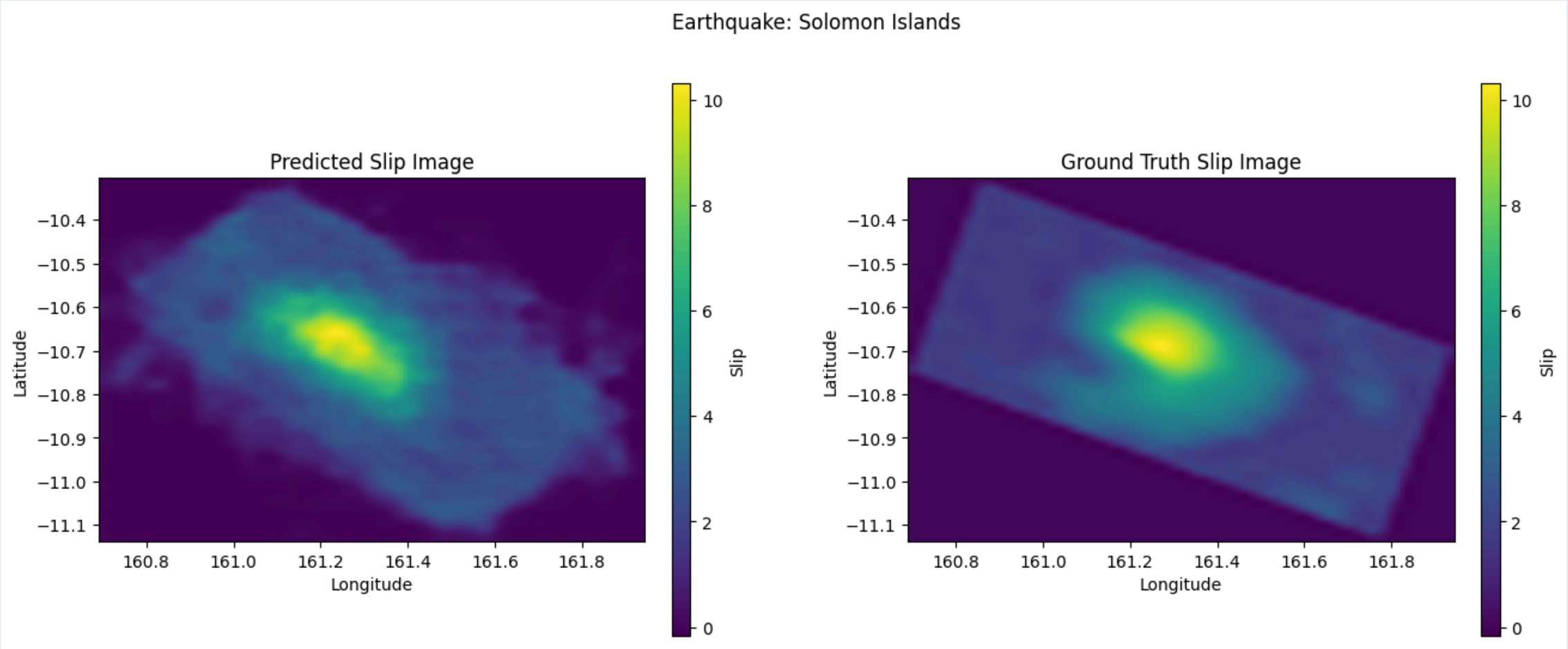
EVALUATION METRICS

Metric	Value
Earth Mover's Distance (emd)	0.0323
Jensen-Shannon Divergence (jsd)	0.1687
Correlation Coefficient (Corr)	0.7029
Structural Similarity Index (SSIM)	0.2614
Peak Signal-to-Noise Ratio (PSNR) [dB]	12.1598
Histogram Correlation (HistCorr)	0.8865
2D PSD Correlation (PSD_Corr_2D)	0.9309
Radial PSD Correlation (PSD_Corr_Radial)	0.9596

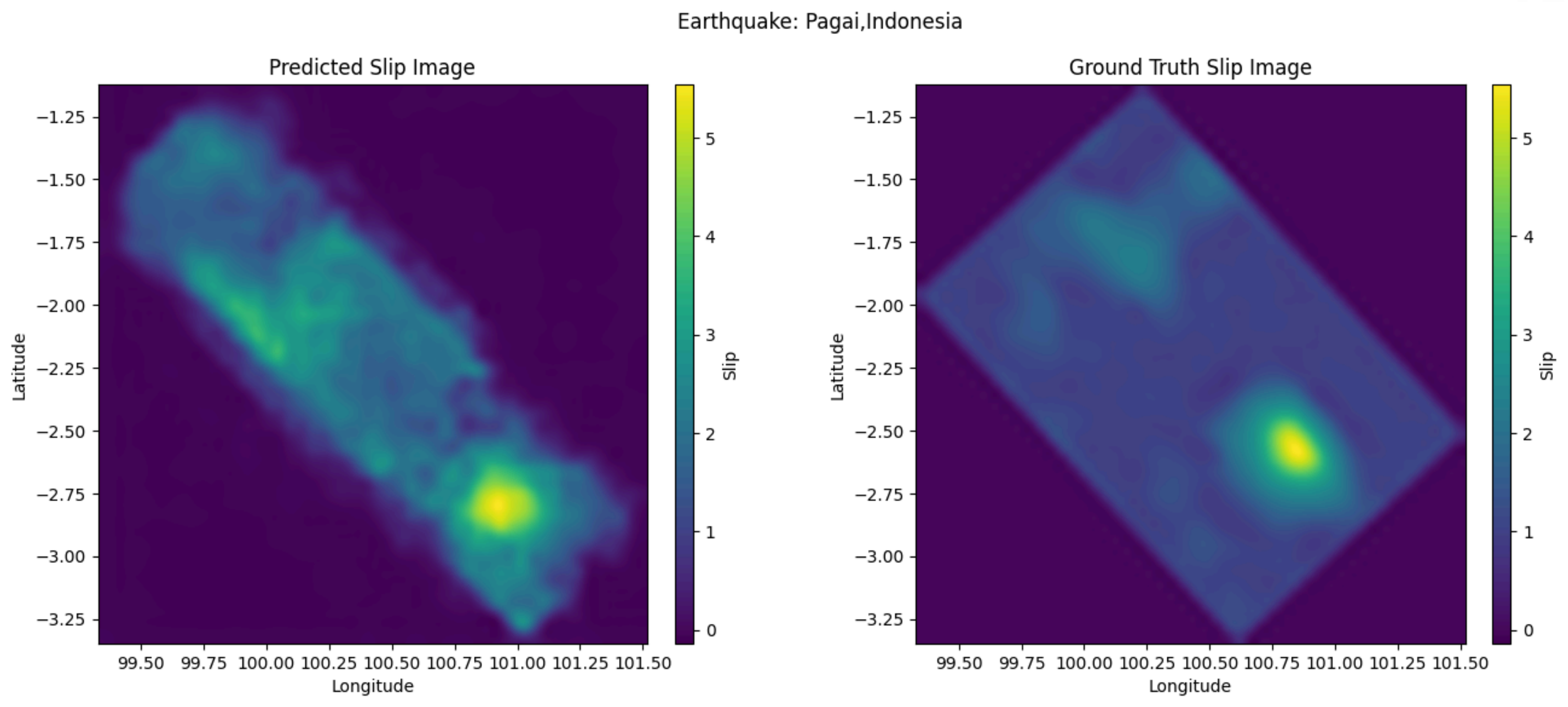
PREDICTIONS



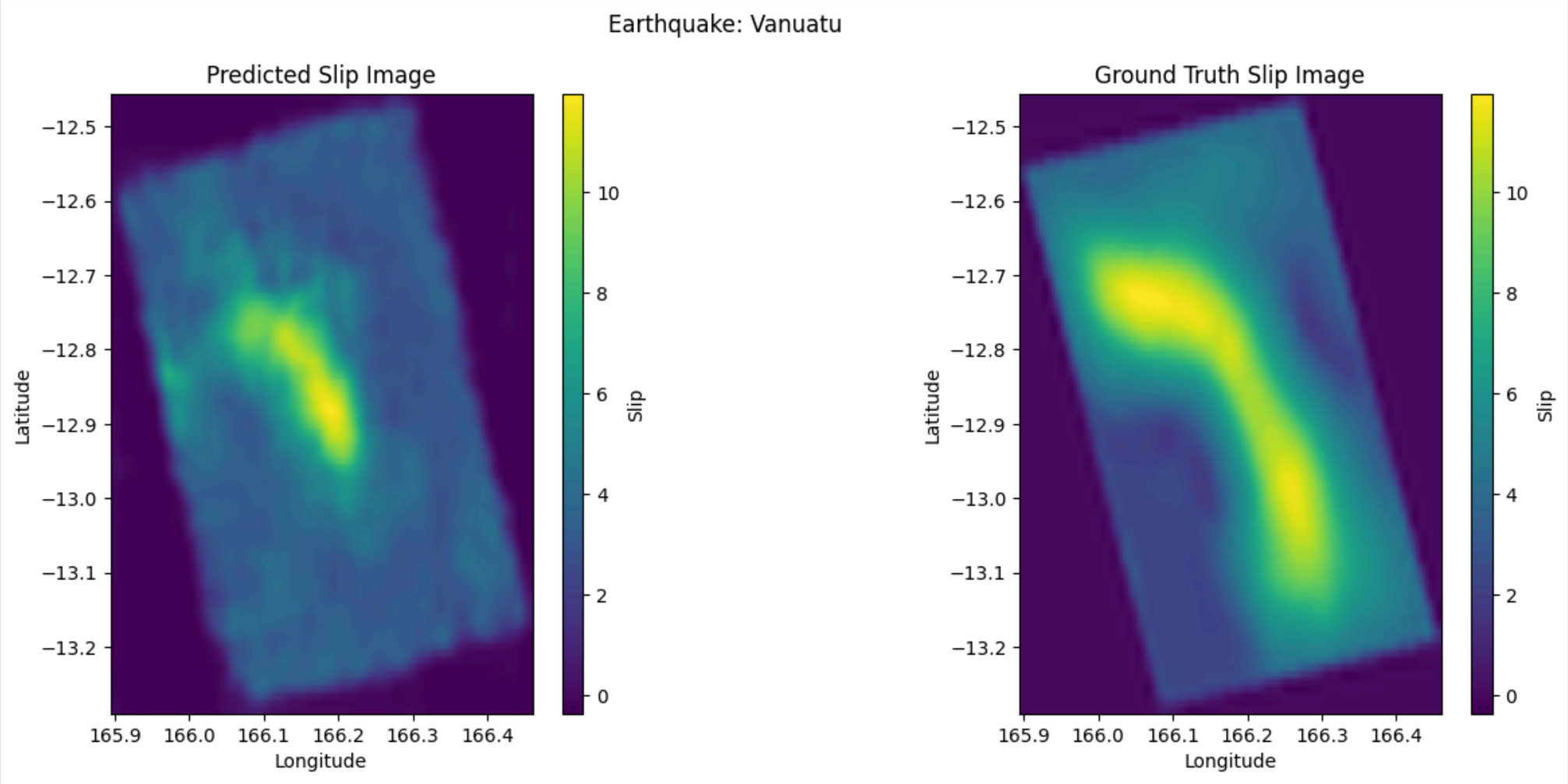
PREDICTIONS



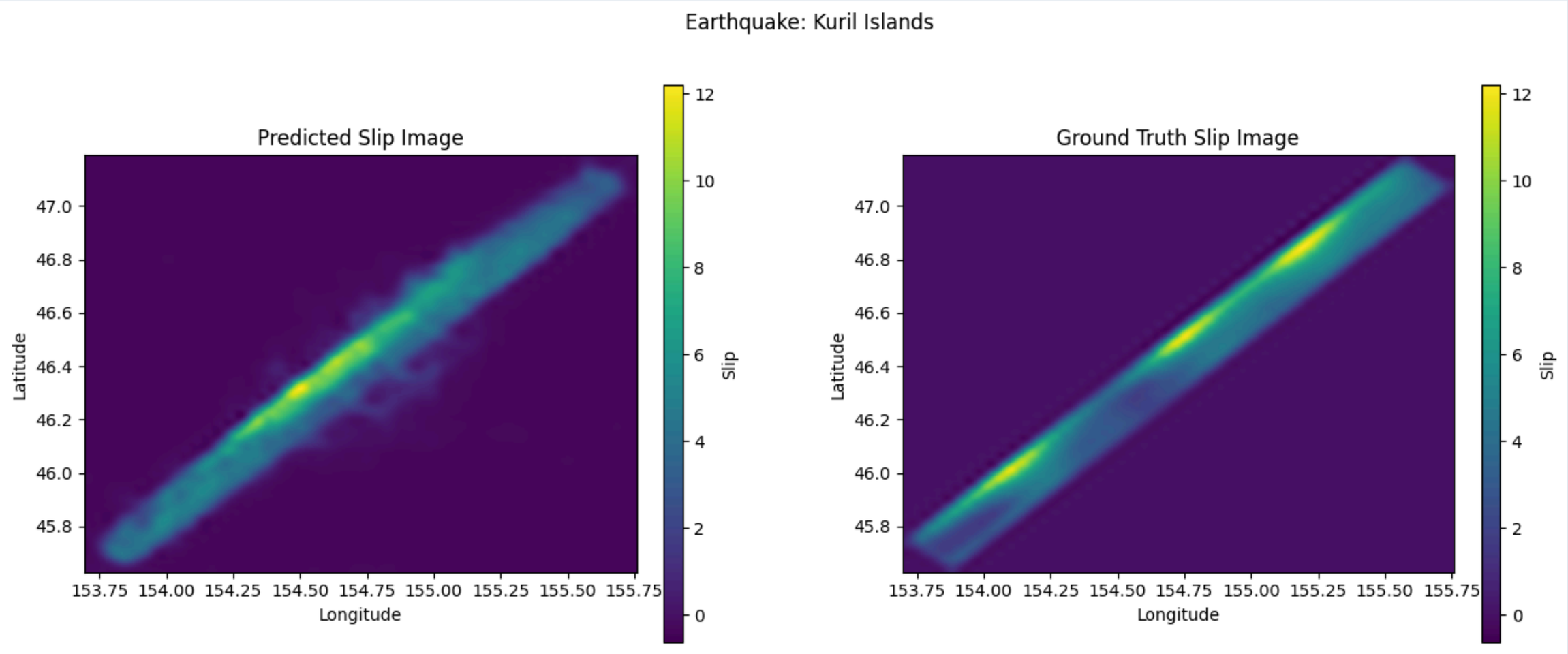
PREDICTIONS



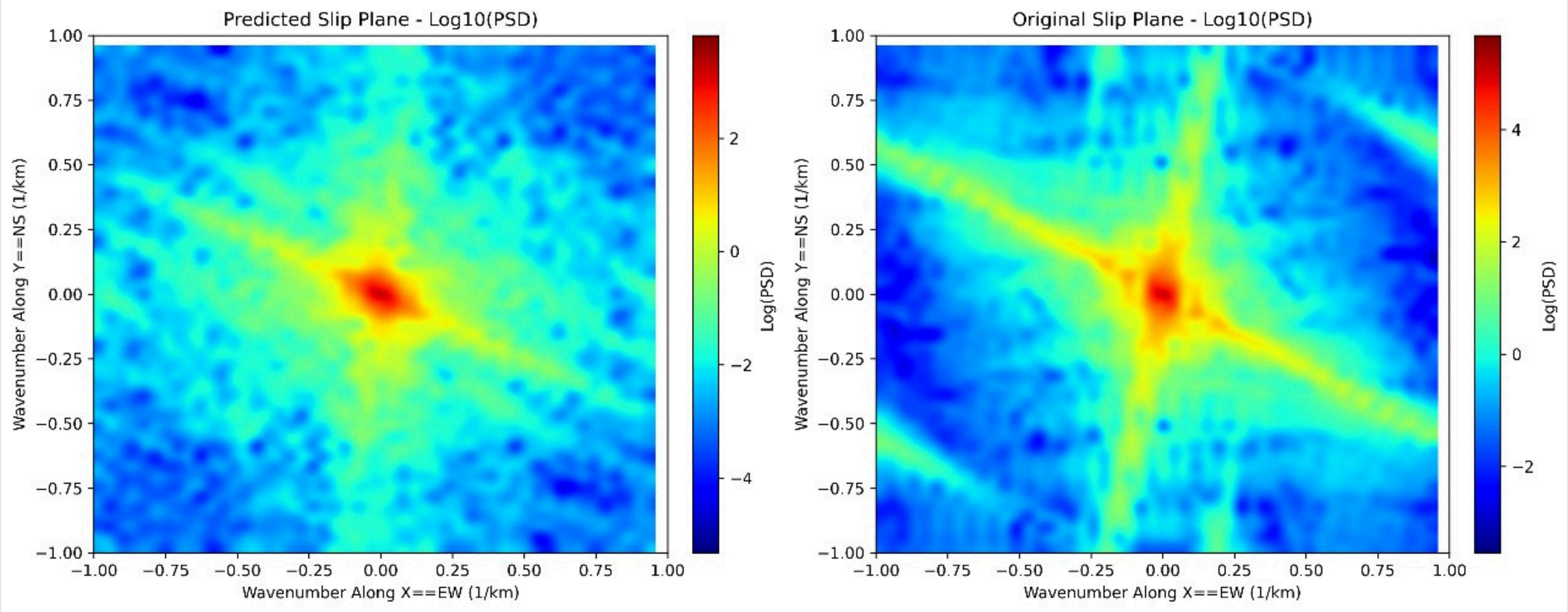
PREDICTIONS



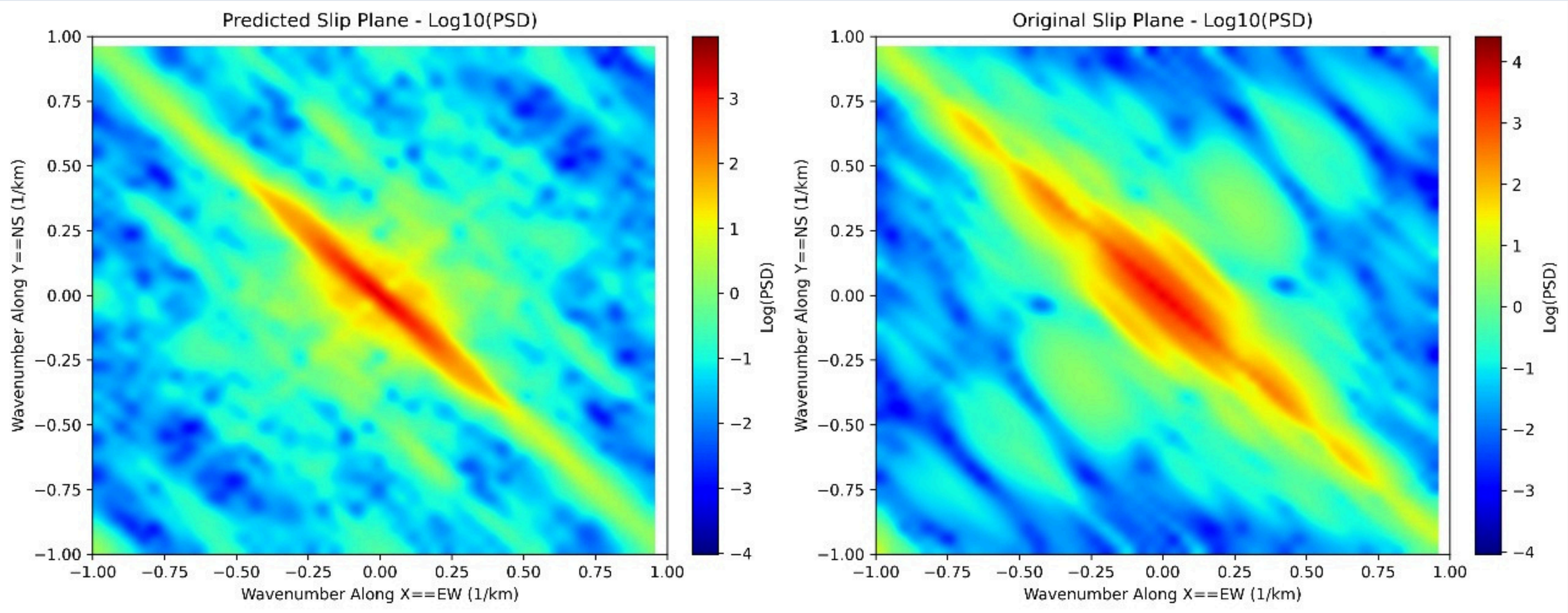
PREDICTIONS



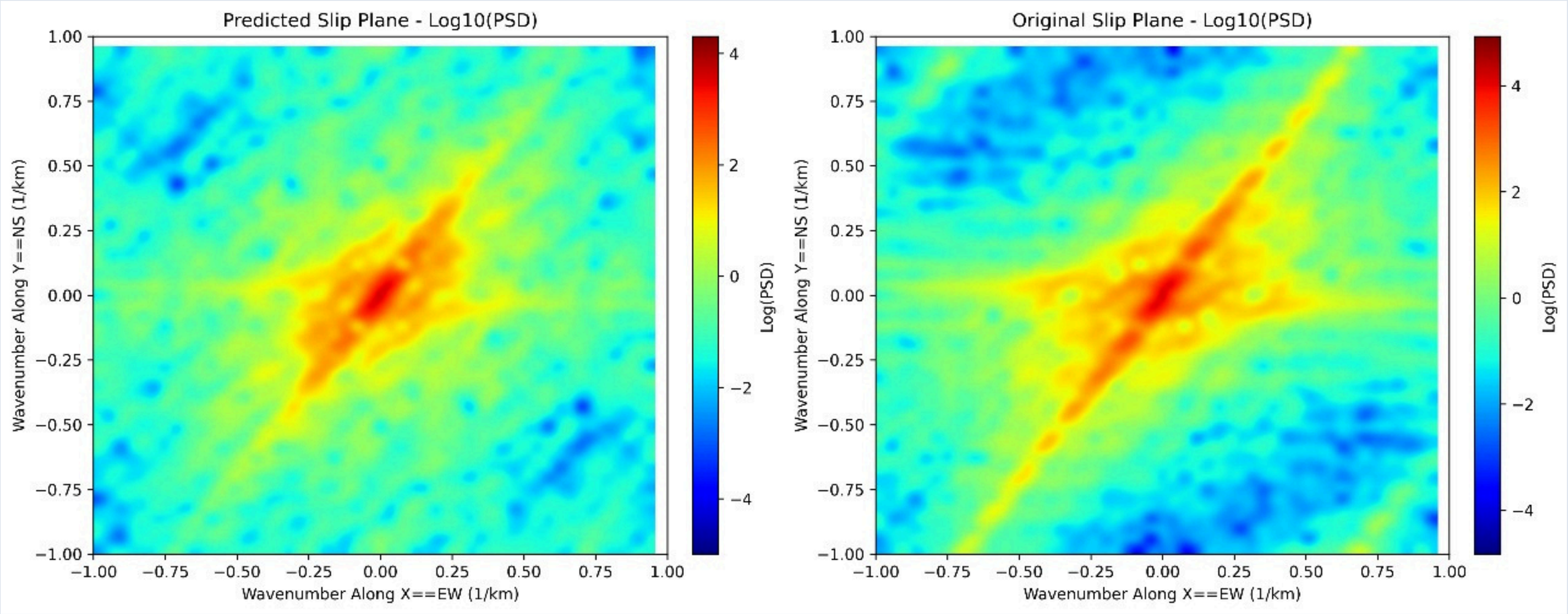
PSD PLOTS


Predicted Slip Plane - Log10(PSD)
Wavenumber Along Y==NS (1/km)
Wavenumber Along X==EW (1/km)
Log(PSD)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2
0
−2
−4
Original Slip Plane - Log10(PSD)
Wavenumber Along Y==NS (1/km)
Wavenumber Along X==EW (1/km)
Log(PSD)
4
2
0
−2

PSD PLOTS


Predicted Slip Plane - Log10(PSD)
Wavenumber Along Y==NS (1/km)
Wavenumber Along X==EW (1/km)
Log(PSD)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
3
2
1
0
−1
−2
−3
−4
Original Slip Plane - Log10(PSD)
Wavenumber Along Y==NS (1/km)
Wavenumber Along X==EW (1/km)
Log(PSD)
4
3
2
1
0
−1
−2
−3
−4

PSD PLOTS


Predicted Slip Plane - Log10(PSD)
Wavenumber Along Y==NS (1/km)
Wavenumber Along X==EW (1/km)
Log(PSD)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
4
2
0
−2
−4
Original Slip Plane - Log10(PSD)
Wavenumber Along Y==NS (1/km)
Wavenumber Along X==EW (1/km)
Log(PSD)

CONCLUSION

- We developed a fast generative framework that predicts 2D slip distributions directly from finite-source parameters.
- The VQ-VAE learns realistic slip patterns, while the parameter-mapping network generates physically consistent rupture scenarios.
- Results show strong spectral and spatial agreement with real SRCMOD slip maps.
- The method bypasses expensive inversion and simulation workflows.
- This approach enables the rapid generation of scenarios for seismic hazard, stress-transfer studies, and research applications.

FUTURE SCOPE

- Use a better interpolation technique than bicubic interpolation.
- Incorporate more features to check if they can affect the model's accuracy
- Hyper-parameter tune it more properly with multiple trials.

THANK YOU